Manufacting / Servce I

Time Limit

2min

Joint Connection

Watch the video

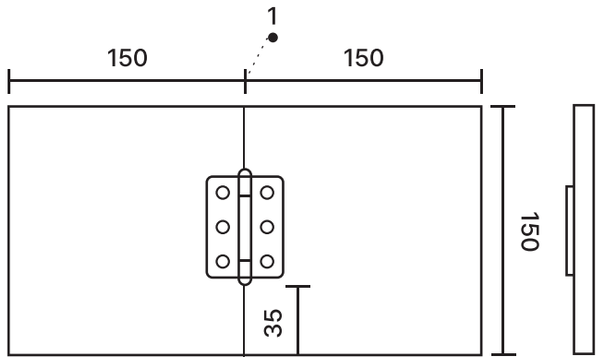
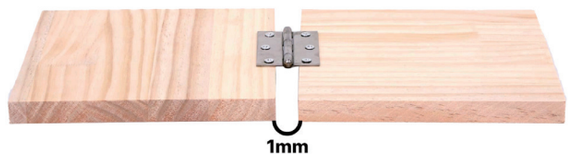
Job Category	Manufacting / Servce I	Subject	Joint Connection

I Requirements

– Bind and stick the plywood to the indicated joint position accurately by using designated toold and parts and dismantle.

II Diagram & Requirements

1) Diagram

Front, Side View	Work Method
	

Job Category	Manufacting / Servce I	Subject	Joint Connection

2) Requirements

① Pay attention to the position and number of nails and use the tool to attach the provided hinges to the designatedlocation within the specified time limit as per the designated drawing

② Please link the hinge and the plates, considering 1mm gap between the plates.

③ Please be conscious of the fact that you will be given the lowest point if the plates couldn't be folded after linking thehinge.

④ Please link the hinge and the plates, considering 1mm gap between the plates.

⑤ Be aware of the fact that you will be given the lowest point if the corresponding scale in more than 20mm.

Ⅲ Equipment & Materials

Equipment & Materials		Spec.	Unit	QTY	Remakrs
	Workbench	· 1,400 * 1,000 * 700 mm	PCS	1	For 1 person
	Plywood	· 150 * 150 * 18 mm · Reusable for up to 4 people	Sheet	2	For 1 person
	Screw	· Ø 3-15mm	PCS	6	For 1 person
	Driver	· Electric driver	PCS	1	
	Hinge	· 3"	PCS	1	

Time Limit

2M 30S

Measure and Cut

Watch the video

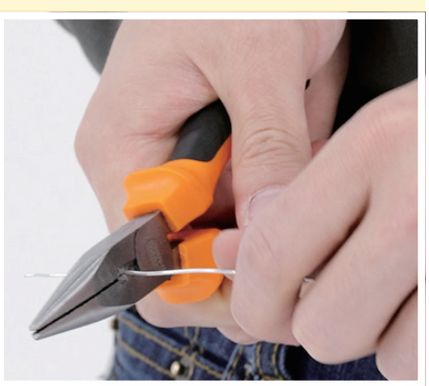
Job Category	Manufacting / Servce I	Subject	Measure and Cut

I Requirements

– Measure weight and length of the given substances, and cut the given material to the desired length.

II Diagram & Requirements

1) Diagram

Meausre Weight	Measure Length	Cutting
		

2) Requirements

⓪ (Pre) After confirming the level of the scale, adjust the zero adjustment screw to align the scale reading to 0.

① Measure the weight of a given item with a scale and record it on paper.

② Measure the length of width, length, inner diameter, outer diameter of the given item and recored it on paper.

③ Cut the wire using a ruler to the length indicated by instructor.

Job Category	Manufacting / Servce I	Subject	Measure and Cut

Ⅲ Equipment & Materials

Equipment & Materials		Spec.	Unit	QTY	Remakrs
	Scale	· Unit for 100g / 30kg	PCS	1	For 1 person
	Items	· Weight Measurement Goods (Box) 5–10 kg · Have a number of items of various weights	PCS	1	Select by the supervisor
	Nipper	· Regular Nipper	PCS	1	For 1 person
	Wires, Ruler	· Minimum length 15cm (or more)	PCS	3/1	(Wire) 3 pieces per person, (Ruler) 1 piece per person
	Component	· Items for measuring legnth	PCS	1	Available in a variety of sizes
	Vernier Calipers	· Digital Vernier Calipers	PCS	1	1 piece per person (If not locally available, replace with a ruler)

Time Limit

7min

Pipe Volting

Watch the video

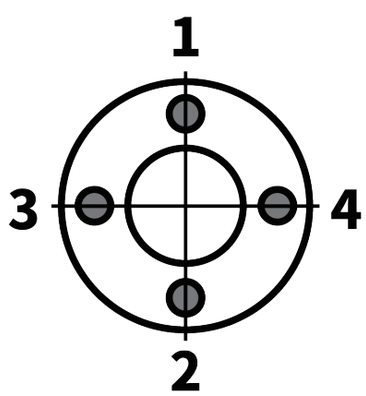
Job Category	Manufacting / Servce I	Subject	Pipe Volting

I Requirements

- Use the tool to join the piping flanges.

II Diagram & Requirements

1) Diagram

Assemble Sequence	Example
	
Result	**Result (Side)**
	

Job Category	Manufacting / Servce I	Subject	Pipe Volting

2) Requirements

① Assemble the flange using the provided materials and tools, and obtain confirmation from the supervising officer.

<order of assembly>

1. Insert the gasket between the flanges and align it.
2. Lightly tighten the bolts, then check the offset between the two flanges.
3. Using a spanner, tighten the bolts and nuts in a diagonal sequence.
4. During the final tightening, confirm that there is no offset on the flange to complete the task.

② After the supervisor's confirmation, disassemble the assembled flange and gasket within the time limit according toinstructions.

Job Category	Manufacting / Servce I	Subject	Pipe Volting

Ⅲ Equipment & Materials

Equipment & Materials		Spec.	Unit	QTY	Remakrs
	Flange	· 5K 50A	PCS	2	For 1 person
	Gasket	· 5K 50A	PCS	1	For 1 person
	Hex Bolt	· M12, 45mm	PCS	4	For 1 person
	Hexagon Nut	· M12	PCS	4	For 1 person
	Washer	· M12	PCS	8	For 1 person
	Spanner	· 19mm	PCS	2	For 1 person
	Vernier Calipers	· Digital Vernier Calipers · Replace with a ruler if local supply is not available	PCS	1	For 1 person

Time Limit

2min

Using Grinders

Watch the video

Job Category	Manufacting / Servce I	Subject	Using grinders

I Requirements

– Understand how to combine / disassemble and use the grinder.

II Diagram & Requirements

1) Diagram



Job Category	Manufacting / Servce I	Subject	Using grinders

2) Requirements

① Attach the grinder handle.

② Attach the guard cover to the grinder and use a hex wrench to tighten the screw completely.

③ Insert the flange(만조), disc, and clamping nut in that order.

④ Press the spindle lock on the grinder to lock the rotating shaft.

⑤ Turn on the grinder, perform a usage simulation, then turn off the grinder.

⑥ After the supervisor's confirmation, if instructed, press the spindle lock and disassemble in the reverse order of assembly.

Ⅲ Equipment & Materials

Equipment & Materials	Spec.	Unit	QTY	Remakrs
 Grinder Set	· 4inch Grinder · Handle · Guard Cover · Disc · Clamping Nut · Grinder Spanner · Hex Wrench	SET	1	For 1 person

Time Limit

50sec

Product Packaging

Watch the video

Job Category	Manufacting / Servce I	Subject	Product Packaging

I Requirements

– Classify and pakage products for delivery properly.

II Diagram & Requirements

1) Diagram

Before Categorization	**After categorization**	After Packaging
		

Job Category	Manufacting / Servce I	Subject	Product Packaging

2) Requirements

⓪ (Pre) Put all the beads in the integration box.

① Sort the beads by color and transfer them to the appropriate boxes.

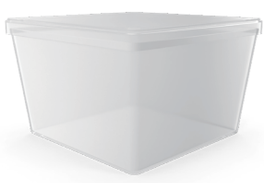
② Classify the 10 or more beads presented according to color and number, pack them in a zipper bag.

③ When all the packaging is done, raise your hand and check with the instructor.

④ After checking, put all the beads back in the integration box.

Job Category	Manufacting / Servce I	Subject	Product Packaging

Ⅲ Equipment & Materials

Equipment & Materials	Spec.	Unit	QTY	Remakrs
 Box	· 190mm*130mm*102mm or more (for integration)40a size of 40 beads	PCS	1	For 1 person
	· 170mm*115mm*89mm or more (for classification) · 10a size of 10 beads	PCS	3	
 Beads	· 31 set of 3 colors, a total of 3 sets · 10 beads in each color, a total of 30 beads, 1 set(Red and green must be included) · Beads size : 15~20mm	SET	1	For 1 person
Zipper Bag	· 10*15cm 10*15 cm or more	PCS	1	For 1 person

Time Limit

50sec

Pallet Loading

Watch the video

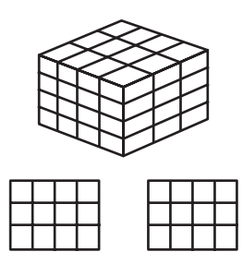
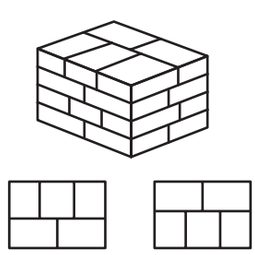
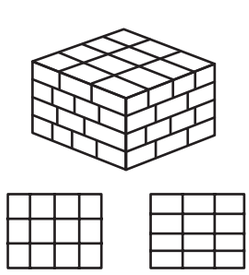
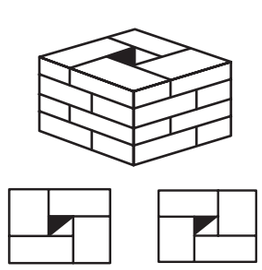
Job Category	Manufacting / Servce I	Subject	Pallet Loading

I Requirements

– Stack boxes in various forms on pallets loaced on the floor.

II Diagram & Requirements

1) Diagram

Block Type	Brick Type
	
Alternate Row Type	**Pinwheel Type**
	

Job Category	Manufacting / Servce I	Subject	Pallet Loading

2) Requirements

① Carry out the box loading operation on the pallet using the loading insturcted method.

❶ Block type : The most common lodading method, stacking boxes in the same shpae and direction.

❷ Brick type : Stack loading products in combination horizontally and vertically at the same stage, and switch 180 degreesat the next stage to stack them at an intersection.

❸ Alternate row : Stacking products by changing the direction of 90 degrees at an inersection.

❹ Pinwheel : A windmill type cargo loading method, similar to a brick type method, in which the steps are stacked in anintersecting form.

❖ Caution : You can transport a maximum of 3 boxes at a time.

② After the supervisor's confirmation, organize the boxes in a block shape according to the instructions.

Job Category	Manufacting / Servce I	Subject	Pallet Loading

Ⅲ Equipment & Materials

Equipment & Materials		Spec.	Unit	QTY	Remakrs
	Box	· 300mm*250mm*210mm(Standard) · Paper has a high damage rate, so it is recommended to usematerials such as plastic	PCS	36	For 1 person
	Pallet	· 1,100mm*1,100mm*150mm(Standard)	PCS	1	For 1 person